

STUDENT PERFORMANCE ANALYSIS REPORT

Data Analytics Project Report

Prepared By:

Reha Arif

Role: Data Analyst

Project Type: Exploratory Data Analysis (EDA) — Academic Performance

2. Executive Summary

This report presents a comprehensive analysis of student academic performance based on a dataset of 300 students across eight class sections (Grades 9 through 12) and five core subjects, totaling 1,500 student-subject records. The analysis was conducted to identify the key behavioral and academic factors that most strongly influence student outcomes, with the objective of informing evidence-based decisions by teachers and academic administration. Following a structured data-cleaning process to address missing values, duplicate records, and invalid entries, exploratory and statistical analysis revealed that attendance percentage and daily study hours are the two strongest predictors of academic performance, each showing a strong positive correlation with marks ($r \approx 0.72$ and $r \approx 0.71$, respectively). Students in the top attendance and study-hours quartiles outperformed their bottom-quartile peers by approximately 28 marks on average, while subject-level analysis identified Mathematics as the most challenging subject and English as the strongest-performing subject overall. The findings support a targeted intervention strategy: prioritizing attendance monitoring, structured study-habit programs, and early identification of at-risk students to improve institution-wide academic outcomes.

3. Introduction

3.1 Background

Academic institutions increasingly rely on data-driven insight to understand the drivers of student success and to allocate support resources effectively. Marks alone provide a limited view of performance; without visibility into contributing factors such as attendance and study habits, administrators are left reacting to poor outcomes rather than preventing them. This project was undertaken to convert raw academic records into a structured, evidence-based understanding of student performance.

3.2 Objective

The primary objective of this analysis is to identify the key factors that influence student academic performance — specifically attendance percentage and study hours — and to quantify their relationship with marks across subjects and class groups. A secondary objective is to surface actionable, specific recommendations that academic administration and teaching staff can implement to improve overall results.

3.3 Scope of Analysis

The scope of this analysis covers the full student dataset spanning Grades 9 through 12 (eight sections) and five subjects: Mathematics, Science, English, History, and Computer Science. The analysis includes data cleaning, descriptive statistics, correlation analysis, comparative performance analysis (by subject,

class, and gender), and identification of at-risk students. The report does not extend to causal experimentation (e.g., controlled interventions) — findings are correlational and are intended to guide, not replace, professional academic judgment.

4. Data Overview

The dataset used for this analysis is structured at the student-subject level, meaning each row represents one student's performance in one subject. This structure enables subject-level granularity while still allowing aggregation to the student level for overall performance metrics.

4.1 Dataset Summary

Metric	Value
Total records (student-subject rows)	1,500
Unique students	300
Class sections covered	8 (Grades 9-12, Sections A & B)
Subjects covered	5 (Mathematics, Science, English, History, Computer Science)
Time period	Current academic term

4.2 Key Fields

Field	Description
student_id	Unique identifier for each student
name	Student full name
gender	Male / Female
class	Grade and section (e.g., 10-A)
subject	Subject associated with the marks record
marks	Marks scored, on a 0-100 scale
attendance_percentage	Overall attendance percentage
study_hours	Average daily self-study hours

5. Methodology

5.1 Data Cleaning

Prior to analysis, the raw dataset was systematically cleaned to ensure accuracy and reliability of findings:

- Missing values: 31 missing marks, 31 missing attendance values, and 30 missing study-hour values were identified and imputed using subject-wise (marks) and overall (attendance, study hours) median/average values.
- Duplicate records: 15 exact duplicate rows were identified and removed.
- Invalid values: Marks and attendance values outside the valid 0-100 range (data-entry errors) were corrected or capped to valid bounds.
- Data type consistency: Fields such as study_hours, which contained mixed numeric/text entries, were standardized to a consistent numeric type.
- Text standardization: Categorical fields (e.g., gender) were trimmed and normalized for consistent casing prior to grouping and aggregation.

5.2 Analytical Approach

The analysis followed a structured exploratory data analysis (EDA) workflow: descriptive statistics were computed for all numeric fields; performance was aggregated and compared across subjects, classes, and gender; Pearson correlation coefficients were calculated to quantify the strength of the relationship between marks and both attendance and study hours; students were segmented into quartiles to compare high- versus low-attendance and high- versus low-study-hours groups; and a combined-risk filter was applied to identify students with both low attendance and low marks.

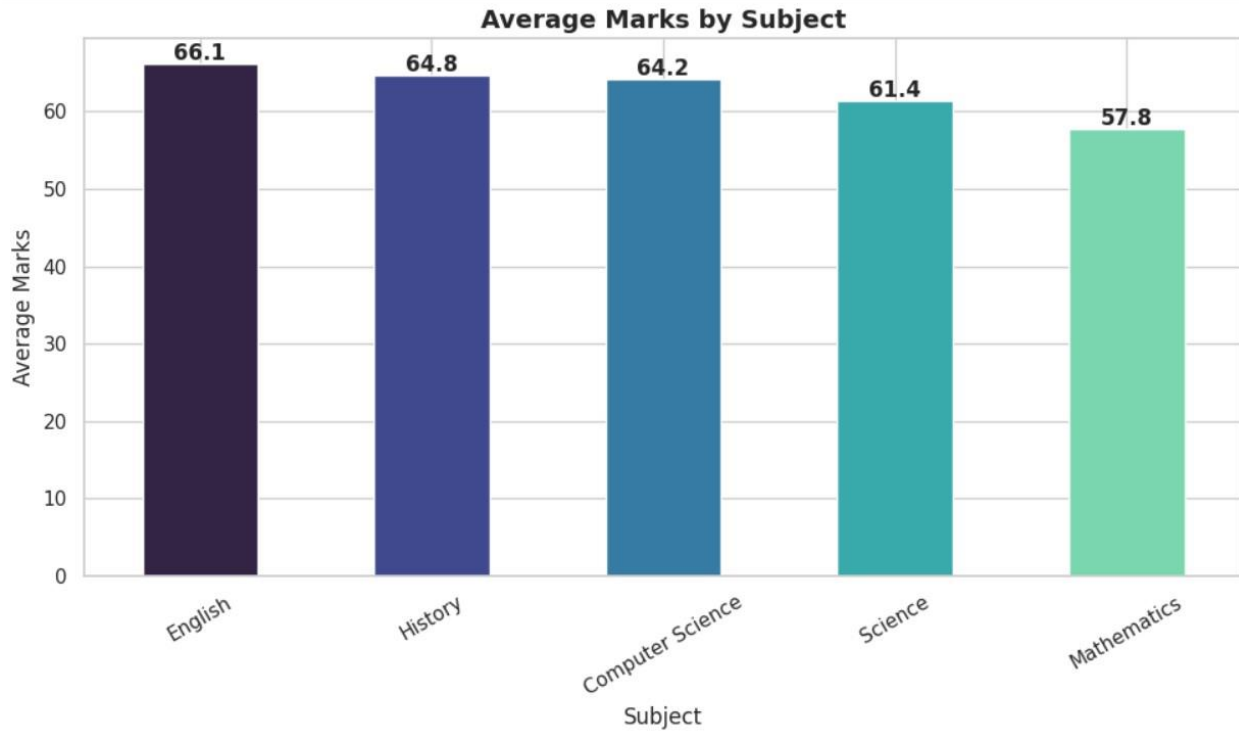
5.3 Tools Used

Tool	Purpose
Python (pandas, numpy)	Data cleaning, transformation, and statistical computation
Matplotlib & Seaborn	Data visualization (charts, plots, heatmaps)
Jupyter Notebook	Analysis environment and documentation
SQL	Structured querying, ranking, and grouped aggregation

6. Key Findings

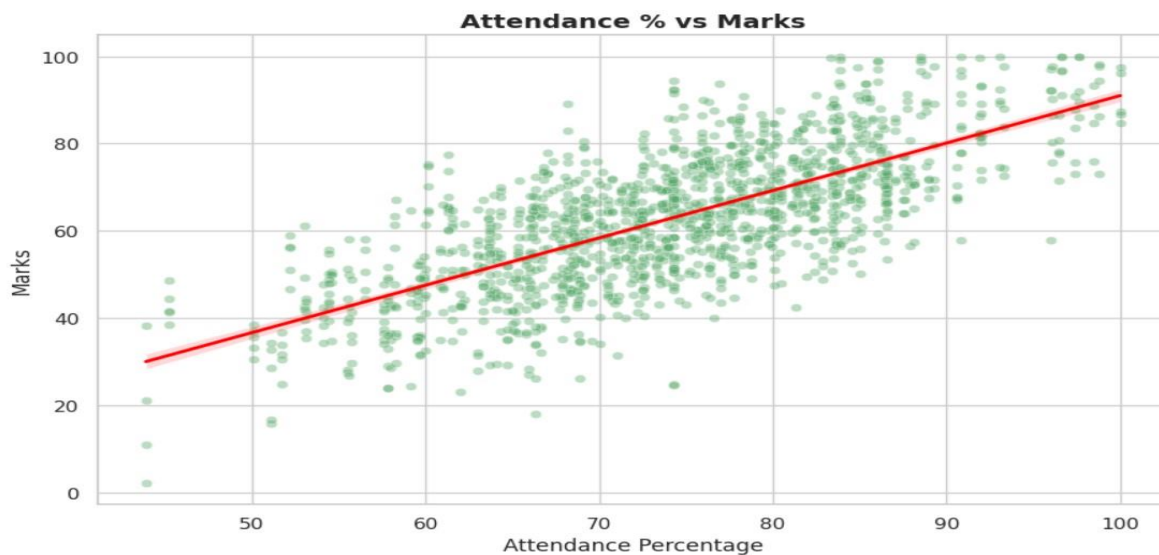
6.1 Subject-Wise and Overall Performance Trends

Average performance varies meaningfully across subjects. English recorded the highest average score (66.1), followed closely by History (64.8) and Computer Science (64.2), while Science (61.4) and particularly Mathematics (57.8) recorded the lowest averages. The overall average across all subjects and students stands at 62.8 marks, with an institution-wide pass rate of 92.9% (using a 40-mark passing threshold).



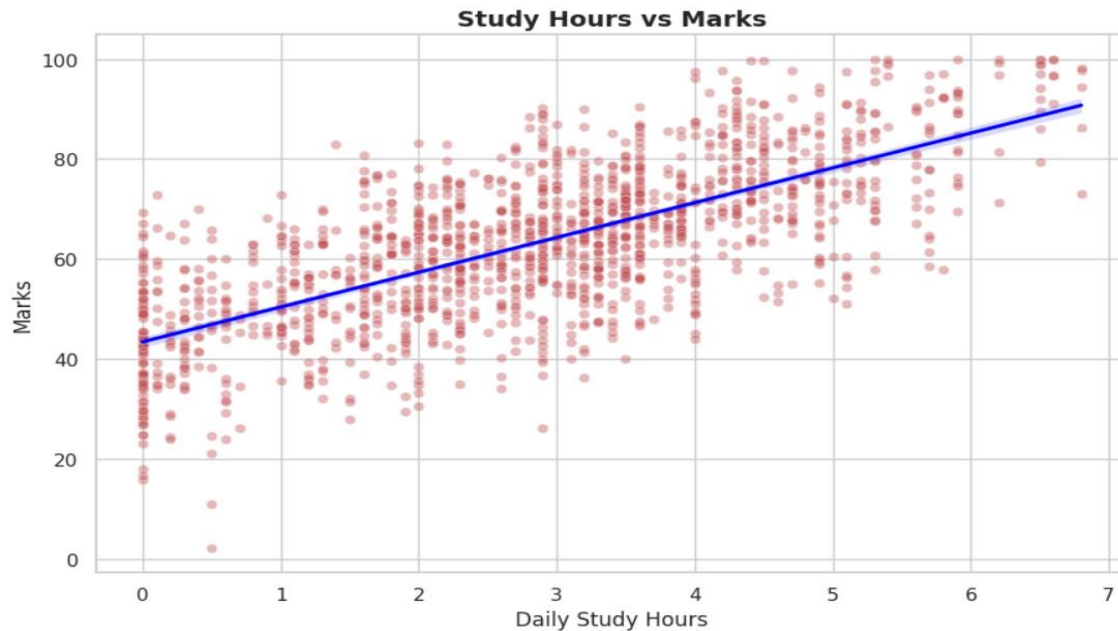
6.2 Attendance vs. Performance Relationship

Attendance percentage shows a strong positive correlation with marks ($r = 0.72$), making it the single strongest measurable predictor of academic performance in this dataset. Students in the top attendance quartile averaged 76.8 marks, compared to 48.3 marks for students in the bottom attendance quartile - a gap of approximately 28.5 marks. When grouped into attendance bands, average marks rise consistently from the lowest band (below 60% attendance) to the highest (90-100% attendance), confirming a near-linear relationship rather than a threshold effect.



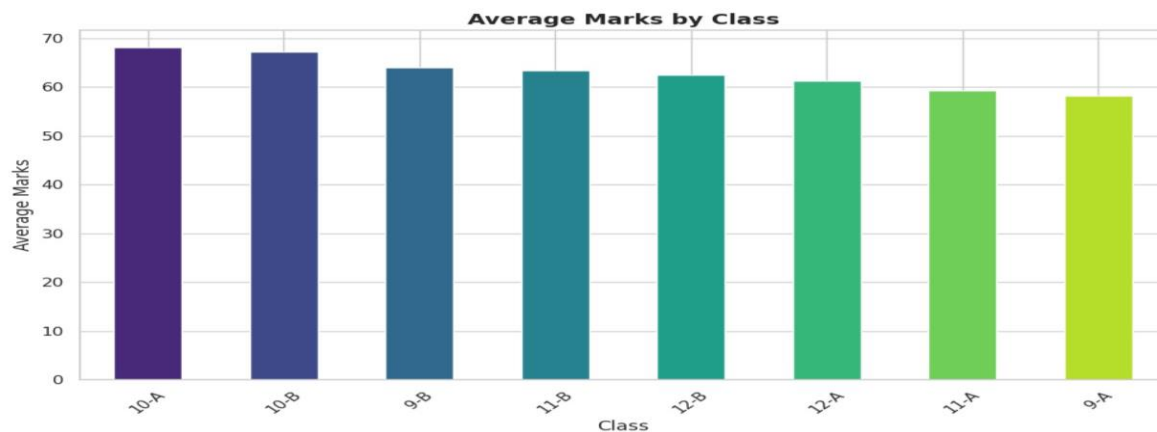
6.3 Study Hours vs. Performance Relationship

Daily study hours show a similarly strong positive correlation with marks ($r = 0.71$). Students in the top study-hours quartile averaged 77.1 marks versus 48.7 marks for the bottom quartile - a gap of approximately 28.4 marks, nearly identical in magnitude to the attendance effect. This indicates that consistent, disciplined self-study is as influential as physical class attendance, and that the two factors likely reinforce one another rather than acting as substitutes.



6.4 Class/Group-Wise Performance Comparison

Performance also varies by class section, though the spread is narrower than the attendance or study-hours effect. Section 10-A recorded the highest class average (68.2), while Section 9-A recorded the lowest (58.2) - a gap of 10.0 marks, roughly one-third the size of the attendance or study-hours gap. This suggests that while class-level factors (teaching quality, peer environment, curriculum pacing) do influence outcomes, individual student habits remain the dominant driver of performance.



6.5 Top 5 and Bottom 5 Performing Students

The following students represent the highest and lowest overall average performers across all subjects:

Rank	Student	Class	Avg. Marks
1 (Top)	Kamran Sheikh	9-B	96.9
2	Hira Yousaf	10-B	96.2
3	Shahzaib Sheikh	10-B	93.7
4	Bushra Awan	10-A	93.5
5	Kamran Butt	9-B	90.4

Rank	Student	Class	Avg. Marks
1 (Bottom)	Ali Farooq	10-B	19.3
2	Hamza Abbasi	11-A	25.6
3	Fatima Hussain	9-B	31.5
4	Ayesha Baig	9-A	34.9
5	Saad Baig	9-A	35.8

Notably, all five bottom-performing students also show below-median attendance and study hours, reinforcing the findings in Sections 6.2 and 6.3.

6.6 Pass/Fail Distribution by Subject and Class

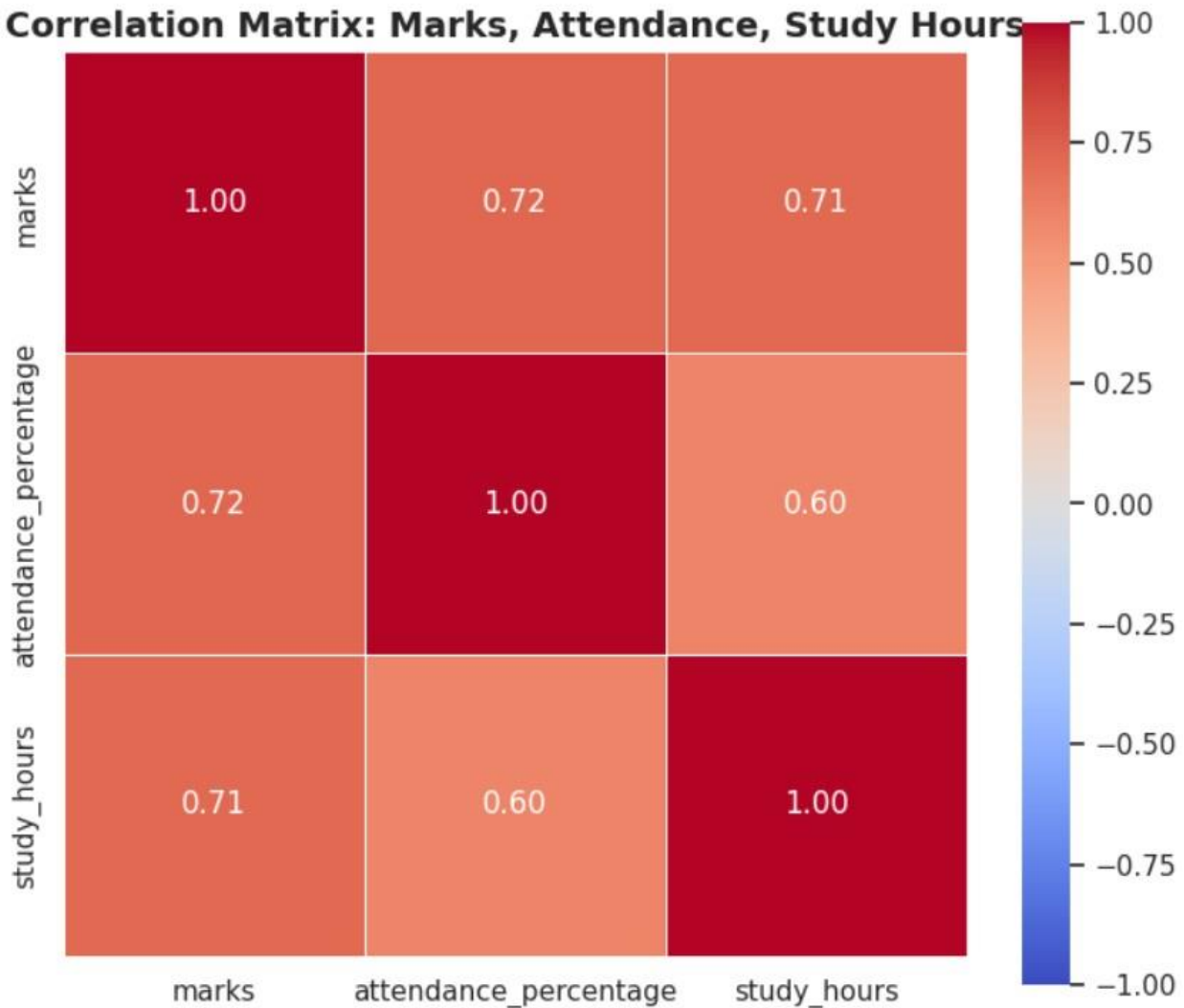
Using a 40-mark passing threshold, pass rates remain high overall but are lowest in Mathematics, consistent with its lower average score:

Subject	Pass Count	Fail Count	Pass Rate
English	285	15	95.0%
History	283	17	94.3%
Computer Science	282	18	94.0%
Science	279	21	93.0%
Mathematics	264	36	88.0%

At the class level, fail counts are highest in Sections 9-A and 12-B, both of which also rank among the lower-performing sections identified in Section 6.4.

6.7 Key Factors Most Affecting Performance

Ranking all measured factors by correlation strength with marks confirms that attendance percentage ($r = 0.72$) and study hours ($r = 0.71$) are, by a wide margin, the most influential measurable factors in this dataset. Gender showed no meaningful performance gap (Female average: 62.3; Male average: 63.4 - a difference of only 1.1 marks), indicating that demographic attributes are far weaker predictors than behavioral ones. Subject and class-section membership introduce moderate variation but remain secondary to individual attendance and study habits.



7. Insights & Business Implications

Insight 1 - Attendance is the single most actionable lever available to administration. With a correlation of 0.72 and a 28.5-mark gap between high- and low-attendance students, attendance is not merely a compliance metric but a leading indicator of academic risk. For administration, this means attendance-tracking systems should be treated as an early-warning tool, not just a record-keeping requirement.

Insight 2 - Study habits are as influential as classroom presence. The near-identical correlation and performance gap for study hours ($r = 0.71$, 28.4-mark gap) indicates that time spent in class is only half the equation. For teachers, this implies that guidance on structured self-study routines deserves equal weight to attendance policy in any improvement plan.

Insight 3 - Mathematics requires targeted curricular attention. As the lowest-performing subject on both average marks (57.8) and pass rate (88.0%), Mathematics stands out as a systemic weak point rather than an isolated case. This signals a need for department-level review - of teaching methods, pacing, or supplemental instruction - rather than student-level intervention alone.

Insight 4 - Performance variation is wide, and averages can mask struggling students. The 77-point spread between the top and bottom student performers shows that a single class average conveys little about the distribution beneath it. For decision-makers, this means performance dashboards should surface distribution and bottom-quartile metrics, not averages alone.

Insight 5 - Class-section differences are real but secondary. The 10-point gap between the best and worst class sections is meaningfully smaller than the attendance or study-hours gap, indicating that resourcing decisions should prioritize student-level support programs over blanket class-level policy changes.

Insight 6 - At-risk students are identifiable before final results. Thirteen students combine both below-40 average marks and below-65% average attendance - a clear, identifiable at-risk cohort. This is significant for administration because it demonstrates that at-risk students can be flagged proactively using existing data, well before end-of-term results are finalized.

Insight 7 - Demographic factors are not meaningful performance drivers. The negligible 1.1-mark difference between genders confirms that performance gaps in this dataset are behavioral (attendance, study habits) rather than demographic - supporting a policy focus on habits and support systems rather than group-based assumptions.

8. Recommendations

8.1 Strengthen Attendance Policy and Monitoring

- Implement automated alerts when a student's attendance falls below 75%, triggering outreach to the student and guardian.
- Introduce a monthly attendance review at the class level to catch section-wide declines early.

8.2 Formalize Study-Habit Support

- Establish a recommended minimum daily study-hour target (e.g., 3+ hours), communicated to students and parents.

- Introduce supervised study halls or structured homework periods for students with historically low study hours.

8.3 Provide Targeted Subject-Level Support

- Launch additional tutoring sessions and revision workshops for Mathematics, given its comparatively low average and pass rate.
- Review Mathematics teaching pace and materials at the department level to identify systemic gaps.

8.4 Build a Proactive At-Risk Student Program

- Use the combined low-attendance and low-marks criterion to generate a standing at-risk student list each grading period.
- Assign counselors or academic mentors to at-risk students for regular check-ins before performance further declines.

8.5 Shift Reporting Toward Distribution, Not Just Averages

- Update academic dashboards to report bottom-quartile performance alongside class averages.
- Track top and bottom performers each term to catch both intervention needs and recognition opportunities.

8.6 Direct Resources Toward Student-Level, Not Just Class-Level, Interventions

- Prioritize individualized support programs over broad class-reassignment policies, given the modest class-level performance gap relative to individual habits.

9. Conclusion

This analysis of 300 students across five subjects and eight class sections confirms that attendance and study habits - not demographic factors or class placement - are the dominant, measurable drivers of academic performance. The consistency of these findings across correlation analysis, quartile comparisons, and at-risk student identification provides a strong evidence base for targeted institutional action. Mathematics emerges as a subject requiring focused curricular attention, while a clearly identifiable group of at-risk students can be proactively supported before results decline further.

The recommendations outlined in this report are intended to be implemented in combination: attendance monitoring and study-habit support address the two strongest performance drivers directly, while subject-level and at-risk interventions target the specific pockets of underperformance identified in the data. As a next step, it is recommended that administration pilot the proposed attendance-alert and at-risk-flagging systems for one academic term, then re-run this analysis to measure the resulting impact on

average marks and pass rates. Extending the dataset in future terms to include additional variables - such as parental involvement, extracurricular participation, or prior-year performance - would further strengthen the predictive value of this analysis.